

WASP-12b

R-1311

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161125 · created 2026-07-31T18:29:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457717.75178 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.01
Transit depth (Rp/Rs)^2	1.66 +/- 0.27 [%]
Orbital Inclination (inc)	86.79 +/- 1.42
Airmass coefficient 1 (a1)	1.156 +/- 0.012
Airmass coefficient 2 (a2)	-0.1402 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	None
Optimal Aperture	3.85
Optimal Annulus	6.57
Transit Duration (day)	0.11531 +/- 0.00098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.01 vs 0.1178 (deviation 1.03σ)
Duration fit vs published	0.11531 d vs 0.125 d (deviation 9.13σ)
Mid-time O–C	0.6 min
Depth SNR	11.9
Residual scatter	0.98 %

Light curve

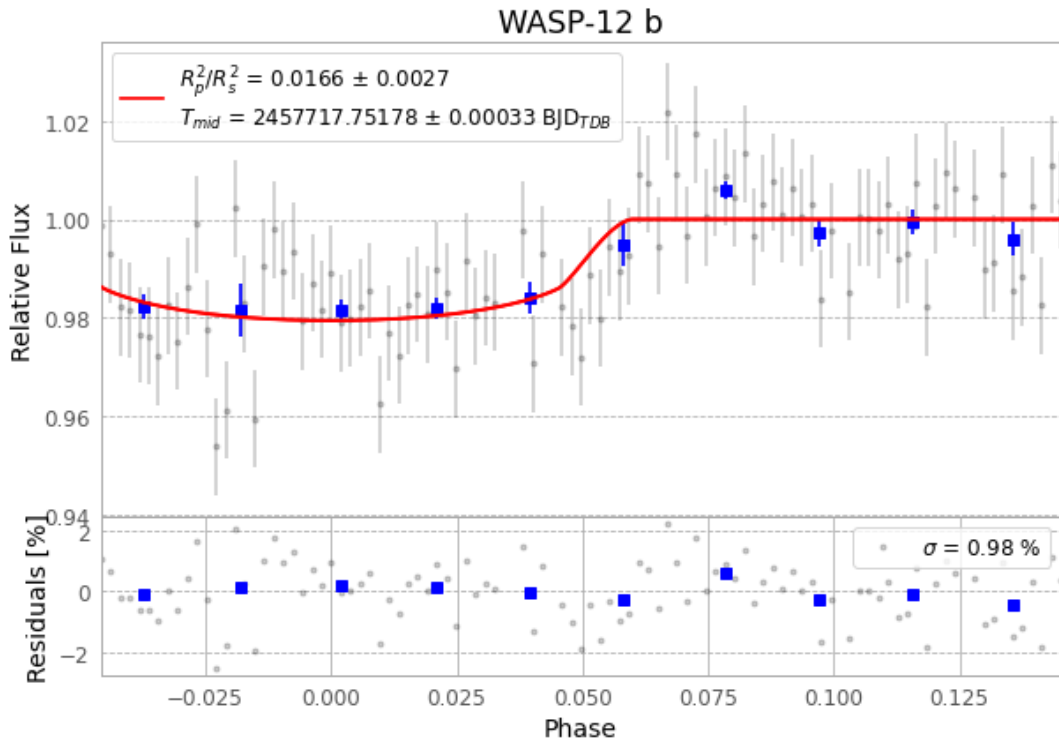


Figure 1 — FinalLightCurve_WASP-12 b_2016-11-24.png (SHA-256 9181af65de06a0f0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17e460e9121b9243...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

99 FITS frames (65 MB), WASP-12161125044212.FITS → WASP-12161125094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161125.log (ccd801e01cc1...), FinalLightCurve_WASP-12 b_2016-11-24.png (9181af65de06...), FinalLightCurve_WASP-12 b_2016-11-24.csv (2caf03276181...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:29Z.